

Nicholas Ketz PhD.

Applied Research Scientist: Artificial Intelligence, Bio-informatics, Mathematical Models, Human Studies

Summary 🏠 Austin, TX, USA ✉️ nick.ketz@gmail.com 🌐 in/nickketz 🌐 nickketz.github.io

I am an applied scientist developing machine learning solutions to advanced research problems. I have more than 10 years of experience in both academic and industrial research labs. I have experience individually contributing, leading small teams and coordinated across multiple departments throughout the entire AI product life cycle.

Relevant Experience

Large Language Modeling	03/2022 - Present
• Developing machine learning solutions for gene therapy and cross-species engineering applications in biotechnology	
Generative AI	01/2020 - 01/2022
• Created AI systems for generating video and images grounded in learned human concepts and semantic understanding	
Model Based Reinforcement Learning	01/2020 - 01/2022
• Developed Agent-based domain adaptation techniques within autonomous driving using model-based reinforcement learning	
Lifelong Deep Learning	09/2018 - 01/2021
• Researched methods for sequential multitask learning in deep neural networks to preserve performance across distribution shifts	
Closed-loop Neural Stimulation	09/2016 - 06/2018
• Developed devices and algorithms for closed-loop neural stimulation in human subjects to enhance cognitive performance	

Education/Employment

Principal AI/ML Scientist <i>Colossal Biosciences/Form Bio</i>	02/2022 - Present
• Leading AI/ML research and development for biotechnology applications	
Research Scientist: Information and Systems Sciences <i>HRL Laboratories</i>	09/2016 - 01/2022
• Conducted research in information and systems sciences with applications in AI	
PhD: Computational Cognitive Neuroscience <i>University of Colorado, Boulder</i>	09/2010 - 09/2016
• Specialized in computational approaches to cognitive neuroscience	
Research Assistant: Davachi Human Memory Lab <i>New York University</i>	09/2007 - 07/2010
• Assisted with research on human memory processes	
BA: Physics, minor: Psychology <i>University of Minnesota, Twin Cities</i>	09/2003 - 06/2007
• Studied physics with a minor in psychology	

Skills

- **Technical:** Deep Learning (CNN, RNN, Transformers, RL), Machine Learning (Optimization, Probabilistic Inference, Unsupervised Learning), Statistics (Bayesian, Parametric/Non-parametric, Time-series), Programming (Python, BASH), Cloud (GCP)
- **AI Lifecycle:** Full-cycle AI development (proof-of-concept to production), cross-team product development, system design/orchestration with containerization, CI/CD pipelines with unit testing and quality assurance
- **Professional:** Team leadership, cross-departmental collaboration (engineering, product, sales), client/stakeholder management, project coordination across multiple research domains and organizations

Select Publications/Patents/White Papers

System and method for semantically grounded video generation	US Patent, 2024
• Generative video AI system prompted by learned human concepts	
Model Interpretability Methods to Predict Gene Therapy Manufacturing Failures <i>White Paper</i>	2022
• Extracting learned motifs from DNA sequence models predictive of gene therapy manufacturing failures	
Independent component selection for automated signal artifact removal <i>US Patent</i>	2020
• Scalable solution for removal of nuisance components in time-series data	
Sliced cramer synaptic consolidation for preserving deeply learned representations <i>ICLR</i>	2019
• Framework for overcoming catastrophic forgetting by preserving the distribution of the network's output at an arbitrary layer	